

2.2 Analytics ↕

Analytics is the scientific process of transforming data in a way that allows us to see hidden patterns and traits, which in turn provides us with the understanding that we need to make better decisions.

Developments in the following three areas have spurred the explosive growth in the use of analytics for decision making:

- technological
- methodological developments
- explosion in computing power and storage capability.

Defining analytics

Analytics enables us to gain insight into the meaning of data, which we might not otherwise detect. The process of analytics incorporates discovering, interpreting and communicating those patterns and traits.

Advances in technology have resulted in a proliferation of data being created. Understandably, organisations are keen to use this data to improve the efficiency, effectiveness and profitability of their operations and to ensure that their customers, clients and users are satisfied with their service. This requires that organisations have a better understanding of:

- their customers'/clients' needs and preferences
- the potential for product improvement (new and existing)
- employee engagement and satisfaction.

□ Watch: What is analytics? [4.38 mins]

This video explains that analytics enables trends to be identified and models to be built, which can allow the probability of an outcome to be predicted. This is useful for making informed decisions in many different situations.

As you watch consider:

- How does the organisation/industry you work for use analytics?
- Why is analytics important to them?



Source: © Educause (2012).

There has been explosive growth in analytics which may involve something as simple as a short report or as complex as optimisations and simulations.

There are four broad categories of analytics:

1. descriptive analytics
2. diagnostic analytics
3. predictive analytics
4. prescriptive analytics.

□ Watch: Types of data analytics – best tools for data analysis [5.33 mins]

This video explains the four different types of analytics and how businesses use them at different stages of development. This is useful as it illustrates that all types of analytics are useful to supporting business decision making.

As you watch consider:

- What type of data analytics does the organisation you work for use?
- Why does the organisation you work for use this/these types of data analytics?



Source: © Precision Health (2020).

A categorisation of analytical methods

Descriptive analytics encompasses a set of analytical tools that describes what has happened in the past.

Examples of analytical data include data queries, standard reports, descriptive statistics, data visualisation, cluster analysis and what-if spreadsheet models. There are four different categories of analytical data. Click on each heading below to learn more about each one.

› **Descriptive analytics**

› **Diagnostic analytics**

› **Predictive analytics**

› **Prescriptive analytics**

□ Reflect

As a postgraduate student, it's important to develop reflective skills, as it leads to more critical thinking. While arguments should be based on empirical findings, it's also important to think as an individual scholar.

Throughout this course, you'll be asked to reflect on your understanding and experience. You should use a dedicated journal or notebook for these reflections, and can use either a hard copy, or an online resource. Your notes are private, and you won't need to submit them, but they will be useful to refer to as you develop your skills and complete your assessments.

In the first activity in this module, we identified a weather app as a way of communicating visual data. Is the data in a weather app predictive or prescriptive? Ensure that you can justify your decision as you may be asked to share your thoughts in the workshop.

□ Activity: The complete list of unicorn companies

Visit [The complete list of unicorn companies](https://instapage.cbinsights.com/research-unicorn-companies) 

(<https://instapage.cbinsights.com/research-unicorn-companies>) (CBInsights, 2023), which provides a list of the world's unicorn companies.

As you scroll through the list, consider and make notes on the following:

- How easy or hard was it for you to understand the web page and the story in the data?
- How would you analyse and visualise the data in the web page?

Explain your reasoning and bring your notes to the workshop.